Zhi-Jun (Jason) Hong

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EDUCATION

Boston University | Boston, MA **Expected Dec 2026**
Master of Science in Data Science | GPA: 3.77/4.0
Relevant Coursework: Machine Learning, Data Engineering, Deep Learning, Data Visualization, Big Data Analytics

National Cheng Kung University (NCKU) | Tainan, Taiwan **Jun 2024**
Bachelor of Business Administration in Statistics | GPA: 3.80/4.0
Interdisciplinary FinTech Program | Fintech Scholarship

SKILLS

Programming Language: Python (Pandas, NumPy, PyTorch, Scikit-learn) , SQL, R
Data Engineering & Databases: ETL Pipeline Development, Feature Engineering, Apache Spark, Kafka, MySQL
Machine Learning: Deep Learning, Neural Networks, Classification / Regression Models, Model Optimization
Data Tools & Platform: Tableau, Git, Jupyter Notebook, Docker, Microsoft Excel

WORK EXPERIENCE

Market Research Intern | Eastern Online Co., Ltd. | Taipei, Taiwan **Jun 2024 - Oct 2024**
- Cleansed and validated 1,000+ ML-generated records, improving data quality for downstream analysis
- Collaborated cross-functionally on model-assisted data pipelines, reducing processing time by 25% through structured data cleaning and validation workflows
- Analyzed product data in Tableau and translated findings into actionable recommendations on market opportunities.

Research Assistant | National Cheng Kung University | Tainan, Taiwan **Sep 2022 - Jul 2023**
- Organized research data and supporting documents, ensuring accuracy and completeness of research submissions
- Created data visualizations in R and structured research outputs in LaTex to communicate analytical findings

DATA SCIENCE PROJECTS

Epidemic Analytics Engine | Big Data Engineering
- Engineered Spark batch pipelines to process four typed Kafka event streams, aggregate epidemiological data, and store structured results in MySQL for downstream analytics
- Designed an ML pipeline for anomaly detection and outbreak prediction, supporting analysis of severity, hospital capacity, vaccination coverage, and epidemic patterns

Medical Case Analytics | Legal Data Analytics
- Collaborated with WBGH and contributed to a regex and fuzzy-matching pipeline by identifying relevant keywords, supporting the classification of unstructured court records into structured medical case categories
- Conducted exploratory data analysis on 10 years of Mass court filings, identifying a 96% decline in ambulance-related cases (2015–2025) while other medical categories remained stable
- Identified concentrated creditor behavior, with Trinity EMS accounting for 60% of ambulance cases and MSPCA Angell representing 91% of veterinary filings

Stroke Risk Prediction | Health Care Machine Learning
- Analyzed 14,000+ patient records across 19 health variables and applied K-Means, GMM, and PCA to identify a 0.68 correlation and segment patients into three distinct risk profiles
- Evaluated five classification models, including Logistic Regression, Random Forest, XGBoost, KNN, and MLP, achieving 0.885 AUC, 80.3% accuracy, and 0.836 F1-score for stroke-risk prediction

Big Data Analysis | Portfolio Optimization & Risk Analysis
- Developed an LSTM model for stock price forecasting and implemented Black–Litterman portfolio optimization in R, benchmarking risk and performance against 30 leading stocks and 100 ETFs
- Evaluated portfolio performance using risk-adjusted metrics, demonstrating that the optimized portfolio outperformed 70% of ETFs and ranked among the top 20 by Sharpe ratio